

CHEYENNE, WY, USA

WMO: 725640

Lat: 41.158N

Lon: 104.807W

Elev: 1863

StdP: 80.85

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24018

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-19.5	-16.1	-23.8	0.5	-15.6	-21.0	0.7	-10.6	17.9	2.7	16.2	2.3	4.0	0	0.709

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	32.1	14.5	30.7	14.2	29.2	14.1	17.2	25.3	16.5	25.0	15.8	24.4	5.2	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.9	13.3	18.9	14.0	12.5	18.6	13.0	11.8	18.2	55.3	25.2	52.9	25.1	50.8	24.3	20.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.9	12.8	11.3	DB	-24.7	34.6	3.5	1.3	-27.3	35.5	-29.3	36.3	-31.3	37.0	-33.9	37.9	(4)
(5)			WB	-25.1	18.8	3.4	0.9	-27.5	19.5	-29.5	20.0	-31.4	20.5	-33.9	21.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	-1.4	-1.5	2.8	5.9	10.9	17.2	21.2	20.0	15.3	8.0	2.5	-1.9
	DBStd	9.66	6.05	6.32	5.72	5.21	4.82	3.88	2.78	2.81	4.60	5.43	6.09	6.27
	HDD10.0	1806	354	323	230	139	48	2	0	0	12	99	229	368
	HDD18.3	3873	612	556	483	373	231	64	9	17	108	319	474	627
	CDD10.0	1190	1	1	5	16	77	218	347	309	172	38	6	0
	CDD18.3	215	0	0	0	0	2	31	97	68	18	0	0	0
	CDH23.3	2969	0	0	0	2	66	515	1195	838	336	17	0	0
	CDH26.7	1017	0	0	0	0	10	168	479	273	86	1	0	0
Wind	WSAvg	5.3	6.2	6.0	5.9	5.9	5.2	4.8	4.3	4.3	4.5	5.2	5.7	6.0
Precipitation	PrecAvg	369	10	10	25	37	58	56	52	42	36	20	14	10
	PrecMax	525	69	32	93	128	152	116	127	169	115	52	63	38
	PrecMin	151	0	0	3	9	3	2	11	4	0	1	0	0
	PrecStd	86	11	8	19	26	33	29	28	28	27	13	14	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.5	16.1	20.7	23.1	28.0	32.9	34.2	32.8	31.3	25.4	20.3	14.6
		MCWB	4.0	5.2	7.2	8.6	12.1	13.8	14.8	14.6	13.8	10.6	7.3	4.3
	2%	DB	11.4	12.9	17.7	20.9	25.1	30.6	32.5	31.1	29.0	23.0	17.2	11.3
		MCWB	2.8	3.6	5.9	8.1	11.3	13.6	14.8	14.2	12.9	9.6	6.0	2.8
	5%	DB	9.2	10.1	15.2	18.6	22.9	28.6	31.2	29.7	27.2	20.8	14.7	9.1
		MCWB	1.9	2.5	4.9	7.2	10.8	13.3	14.6	14.1	12.3	8.8	5.0	2.0
	10%	DB	7.2	7.7	12.6	16.0	20.4	26.5	29.6	28.1	25.1	17.9	11.9	7.1
		MCWB	0.9	1.2	3.9	6.3	10.1	13.4	14.7	14.1	11.8	7.8	3.8	1.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.9	5.6	7.9	10.2	14.7	17.3	18.4	17.8	16.4	11.9	8.0	4.7
		MCDB	12.5	14.9	18.3	19.9	23.0	25.1	26.1	25.6	26.1	21.6	18.4	13.1
	2%	WB	3.4	4.0	6.5	8.9	13.1	16.2	17.5	16.8	15.0	10.7	6.5	3.4
		MCDB	10.2	11.8	16.2	17.8	20.8	24.4	25.5	25.4	23.7	20.1	16.4	10.4
	5%	WB	2.3	2.8	5.4	7.8	11.9	15.2	16.8	16.1	13.9	9.6	5.3	2.3
		MCDB	8.7	9.6	14.2	16.7	19.7	23.6	25.0	24.9	22.0	18.1	13.9	8.7
	10%	WB	1.1	1.5	4.1	6.8	10.9	14.3	16.1	15.5	13.0	8.4	4.0	1.1
		MCDB	6.9	7.3	11.8	15.2	18.6	23.0	24.6	23.8	21.2	16.8	11.3	6.8
Mean Daily Temperature Range	5% DB	MDBR	11.2	11.4	12.6	12.4	12.5	14.3	14.7	14.5	14.4	13.0	12.0	10.9
		MCDBR	13.1	14.2	16.6	17.5	16.7	18.0	17.4	17.2	17.7	17.7	15.8	13.1
		MCWBR	7.3	7.8	8.0	7.9	6.7	6.5	5.4	5.4	6.4	7.8	7.9	7.4
	5% WB	MCDBR	12.4	13.6	15.7	15.9	14.8	15.1	14.3	14.4	14.7	15.8	15.0	12.6
		MCWBR	7.3	7.8	7.8	7.7	6.7	6.7	5.4	5.3	5.9	7.6	7.8	7.4
Clear-Sky Solar Irradiance	taub	0.233	0.240	0.262	0.287	0.308	0.315	0.341	0.336	0.298	0.262	0.245	0.235	
	taud	2.600	2.587	2.536	2.492	2.489	2.492	2.454	2.455	2.553	2.630	2.581	2.568	
	Ebn at Noon	963	1000	1001	988	966	956	927	924	947	958	934	929	
	Edh at Noon	65	77	91	103	107	107	107	110	106	90	73	66	62
All-Sky Solar Radiation	RadAvg	2.35	3.29	4.53	5.39	6.07	7.17	6.90	6.06	5.08	3.57	2.54	1.98	
	RadStd	0.10	0.19	0.33	0.36	0.52	0.36	0.27	0.17	0.29	0.26	0.14	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page